

1. Course number and name

CHE 457 Colloidal and Interfacial Phenomena

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Vivek Sharma

4. Text book, title, author, publisher, year

Lecture notes, assignments and extra reading material form bulk of the class content.

Sample resource materials:

1. Capillarity and Wetting Phenomena: Drops, Bubbles, Pearls, Waves, Pierre-Gilles de Gennes, Francoise Brochard-Wyatt and David Quere, Springer, 2004.
2. The Physics of Foam, Denis Weaire, Stefan Hutzler, Oxford University Press, 1999.
3. Soap Bubbles: Their Colors and Forces which Mold Them, C. V. Boys, Dover 1959
4. The Science of Soap Films and Soap Bubbles, C. Isenberg, Dover 1992
5. Physical Chemistry of Surfaces, 6th edition, Arthur W. Adamson & Alice P. Gast, John Wiley & Sons, 1997.
6. Soap Films: Studies of their Thinning & a Bibliography, Karol J. Mysels, Kozo Shinoja and Stanley Frankel, 1957.
7. The Colloidal Domain, 2nd ed., D. Fennel Evans and Hakan Wennerstrom, Wiley-VCH 1999.
8. Interfacial Transport Processes and Rheology, Darsh Wasan, Edwards and Howard Brenner, Butterworth-Heinemann, 1991
9. Intermolecular and Surface Forces, Jacob Israelachvili, Academic Press, 3rd edition, 2011.
10. Colloidal Dispersions by W. B. Russel, D. A. Saville and W. R. Schowalter, Cambridge University Press, 1988.
11. Froth: The Science of Beer by Mark Denny, Johns Hopkins University Press, 2009.
12. Fizzics: The Science of Bubbles, Droplets and Foams by F. Ronald Young, Johns Hopkins University Press, 2011.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Interfacial phenomena in practice: soap bubbles, emulsions, foams, detergents; surface tension driven flows, Marangoni effect; interfacial rheology; colloids, emulsions, 3D-printing; interfacial thermodynamics, biological interfaces.

Prerequisites or co-requisites

CHE 301, CHE 312.

b. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

1. Demonstrate knowledge of capillarity and wetting (surface tension, contact angle, Laplace pressure, disjoining pressure, interfacial thermodynamics, super hydrophobicity, surfactants, spreading kinetics).
2. Apply the conservation principles and flux laws to model transport processes central to interfacial science and engineering.
3. Apply the physical and mathematical concepts underlying colloidal, emulsion & foam stability, thermodynamics and dynamics to solve new problems “by analogy”.
4. Understand the significant role played by physics and interfacial phenomena in daily life and in typical engineering practice.
5. Develop a fundamental understanding of experimental methods use for measuring surface tension, interfacial rheology & for studying surface-tension driven flows, pattern formation and instabilities.
6. Prepare a well-researched written research proposal and an oral presentation.

a. Explicitly indicate which of the student outcomes are addressed by the course.

	1	2	3	4
(7) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(7) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(7) an ability to communicate effectively with a range of audiences.				X
(7) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(7) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(7) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. Brief list of topics to be covered

1. Introduction: Soap bubbles, soda pop, emulsions, foams, fog, dew, beer froth, shaving foam, whipped cream & detergents, “Ben Franklin stilled the waves” & historical perspective
2. Capillarity & deformable interfaces
3. Physicochemical hydrodynamics
4. Interfacial rheology: fundamental concepts & techniques
5. Interfacial Shear & Dilatational Viscosity/Modulus
6. Thin film hydrodynamics: Viscoelastic interfaces
7. Biological Interfaces
8. Thermodynamics of interfaces
- 9 Liquid crystalline phases
10. Imaging & spectroscopy of interfaces & interfacial phenomena
11. Industrial applications